

prepare_sjsdm_cross_validation_fold.R

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prepare_sjsdm_cross_validation_fold	
<i>Prepare One sjSDM Cross-Validation Fold</i>	

Description

Converts location-assignment row indices to sample identifiers, rebuilds fold-local spatial predictors when needed, and prepares train/test model inputs with training-only response filtering and predictor scaling.

Usage

```
prepare_sjsdm_cross_validation_fold(  
  data_community_matrix = NULL,  
  data_abiotic_wide = NULL,  
  data_coords_projected = NULL,  
  data_sample_ids = NULL,  
  train_indices = NULL,  
  test_indices = NULL,  
  config_model_fitting = NULL,  
  config_data_processing = NULL,  
  repeat_id = NULL,  
  fold_id = NULL,  
  compute_spatial_function = compute_spatial_mev,  
  compute_spatiotemporal_function = compute_spatiotemporal_mev,  
  interpolate_spatial_function = interpolate_mev_to_grid,  
  interpolate_spatiotemporal_function = interpolate_st_mev_to_grid  
)
```

Arguments

- `data_community_matrix`
Raw community matrix with sample identifiers as row names.
- `data_abiotic_wide`
Wide abiotic data frame with 'dataset_name', 'age', and predictors.
- `data_coords_projected`
Projected coordinates used for fold-local MEM construction.
- `data_sample_ids`
Sample metadata containing 'dataset_name' and 'age'. Optional 'sample_id', 'row_index', and 'location_id' columns are derived when absent.
- `train_indices, test_indices`
Integer row indices defining the fold training and held-out samples.
- `config_model_fitting, config_data_processing`
Active project configuration lists.
- `repeat_id, fold_id`
Fold identifiers. They are accepted for runner compatibility and retained by downstream diagnostics.
- `compute_spatial_function, compute_spatiotemporal_function`
Functions used by [prepare_fold_spatial_predictors()] to build MEMs.
- `interpolate_spatial_function, interpolate_spatiotemporal_function`
Functions used by [prepare_fold_spatial_predictors()] to project MEMs.

Details

This helper is the production adapter between location-level CV assignment tables and the generic fold-local preparation functions shared with decomposition diagnostics.

Value

Named list returned by [prepare_model_fold_input()], with optional 'data_spatial_diagnostics' when spatial predictors are enabled.

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